

4. The Itô Formula and the Martingale Representation Theorem

4.1 The 1-dimensional Itô formula

Example 3.1.9 illustrates that the basic definition of Itô integrals is not very useful when we try to evaluate a given integral. This is similar to the situation for ordinary Riemann integrals, where we do not use the basic definition but rather the fundamental theorem of calculus plus the chain rule in the explicit calculations.

In this context, however, we have no differentiation theory, only integration theory. Nevertheless it turns out that it is possible to establish an Itô integral version of the chain rule, called the Itô formula. The Itô formula is, as we will show by examples, very useful for evaluating Itô integrals.

From the example

$$\int_0^t B_s dB_s = \frac{1}{2}B_t^2 - \frac{1}{2}t \quad \text{or} \quad \frac{1}{2}B_t^2 = \frac{1}{2}t + \int_0^t B_s dB_s, \quad (4.1.1)$$

we see that the image of the Itô integral $B_t = \int_0^t dB_s$ by the map $g(x) = \frac{1}{2}x^2$ is not again an Itô integral of the form

$$\int_0^t f(s, \omega) dB_s(\omega)$$

but a combination of a dB_s -and a ds -integral:

$$\frac{1}{2}B_t^2 = \int_0^t \frac{1}{2}ds + \int_0^t B_s dB_s. \quad (4.1.2)$$

It turns out that if we introduce *Itô processes* (also called *stochastic integrals*) as sums of a dB_s -and a ds -integral then this family of integrals is stable under smooth maps. Thus we define

Definition 4.1.1 (1-dimensional Itô processes).

Let B_t be 1-dimensional Brownian motion on $(\Omega, \mathcal{F}, P)$. A (1-dimensional)

Itô process (or stochastic integral) is a stochastic process X_t on $(\Omega, \mathcal{F}, P)$ of the form

$$X_t = X_0 + \int_0^t u(s, \omega) ds + \int_0^t v(s, \omega) dB_s , \quad (4.1.3)$$

where $v \in \mathcal{W}_{\mathcal{H}}$, so that

$$P \left[\int_0^t v(s, \omega)^2 ds < \infty \text{ for all } t \geq 0 \right] = 1 \quad (4.1.4)$$

(see Definition 3.3.2). We also assume that u is $\mathcal{H}_t$ -adapted (where $\mathcal{H}_t$ is as in (ii)', Section 3.3) and

$$P \left[\int_0^t |u(s, \omega)| ds < \infty \text{ for all } t \geq 0 \right] = 1 . \quad (4.1.5)$$

If X_t is an Itô process of the form (4.1.3) the equation (4.1.3) is sometimes written in the shorter differential form

$$dX_t = udt + vdB_t . \quad (4.1.6)$$

For example, (4.1.1) (or (4.1.2)) may be represented by

$$d\left(\frac{1}{2}B_t^2\right) = \frac{1}{2}dt + B_t dB_t .$$

We are now ready to state the first main result in this chapter:

Theorem 4.1.2 (The 1-dimensional Itô formula).

Let X_t be an Itô process given by

$$dX_t = udt + vdB_t .$$

Let $g(t, x) \in C^2([0, \infty) \times \mathbf{R})$ (i.e. g is twice continuously differentiable on $[0, \infty) \times \mathbf{R}$). Then

$$Y_t = g(t, X_t)$$

is again an Itô process, and

$$dY_t = \frac{\partial g}{\partial t}(t, X_t)dt + \frac{\partial g}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2}(t, X_t) \cdot (dX_t)^2 , \quad (4.1.7)$$

where $(dX_t)^2 = (dX_t) \cdot (dX_t)$ is computed according to the rules

$$dt \cdot dt = dt \cdot dB_t = dB_t \cdot dt = 0 , \quad dB_t \cdot dB_t = dt . \quad (4.1.8)$$

Before we prove Itô's formula let us look at some examples.

Example 4.1.3. Let us return to the integral

$$I = \int_0^t B_s dB_s \quad \text{from Chapter 3 .}$$

Choose $X_t = B_t$ and $g(t, x) = \frac{1}{2}x^2$. Then

$$Y_t = g(t, B_t) = \frac{1}{2}B_t^2 .$$

Then by Itô's formula,

$$dY_t = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dB_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} (dB_t)^2 = B_t dB_t + \frac{1}{2} (dB_t)^2 = B_t dB_t + \frac{1}{2} dt .$$

Hence

$$d(\frac{1}{2}B_t^2) = B_t dB_t + \frac{1}{2} dt .$$

In other words,

$$\frac{1}{2}B_t^2 = \int_0^t B_s dB_s + \frac{1}{2}t, \quad \text{as in Chapter 3 .}$$

Example 4.1.4. What is

$$\int_0^t s dB_s ?$$

From classical calculus it seems reasonable that a term of the form tB_t should appear, so we put

$$g(t, x) = tx$$

and

$$Y_t = g(t, B_t) = tB_t .$$

Then by Itô's formula,

$$dY_t = B_t dt + t dB_t + 0$$

i.e.

$$d(tB_t) = B_t dt + t dB_t$$

or

$$tB_t = \int_0^t B_s ds + \int_0^t s dB_s$$

or

$$\int_0^t s dB_s = tB_t - \int_0^t B_s ds ,$$

which is reasonable from an integration-by-parts point of view.

More generally, the same method gives

Theorem 4.1.5 (Integration by parts). *Suppose $f(s, \omega) = f(s)$ only depends on s and that f is continuous and of bounded variation in $[0, t]$. Then*

$$\int_0^t f(s) dB_s = f(t) B_t - \int_0^t B_s df_s .$$

Note that it is crucial for the result to hold that f does not depend on ω . (See Exercise 4.3.)

Sketch of proof of the Itô formula. First observe that if we substitute

$$dX_t = u dt + v dB_t$$

in (4.1.7) and use (4.1.8) we get the equivalent expression

$$\begin{aligned} g(t, X_t) &= g(0, X_0) + \int_0^t \left(\frac{\partial g}{\partial s}(s, X_s) + u_s \frac{\partial g}{\partial x}(s, X_s) + \frac{1}{2} v_s^2 \cdot \frac{\partial^2 g}{\partial x^2}(s, X_s) \right) ds \\ &\quad + \int_0^t v_s \cdot \frac{\partial g}{\partial x}(s, X_s) dB_s \quad \text{where } u_s = u(s, \omega), v_s = v(s, \omega) . \end{aligned} \quad (4.1.9)$$

Note that (4.1.9) is an Itô process in the sense of Definition 4.1.1.

We may assume that g , $\frac{\partial g}{\partial t}$, $\frac{\partial g}{\partial x}$ and $\frac{\partial^2 g}{\partial x^2}$ are bounded, for if (4.1.9) is proved in this case we obtain the general case by approximating by C^2 functions g_n such that g_n , $\frac{\partial g_n}{\partial t}$, $\frac{\partial g_n}{\partial x}$ and $\frac{\partial^2 g_n}{\partial x^2}$ are bounded for each n and converge uniformly on compact subsets of $[0, \infty) \times \mathbf{R}$ to g , $\frac{\partial g}{\partial t}$, $\frac{\partial g}{\partial x}$, $\frac{\partial^2 g}{\partial x^2}$, respectively. (See Exercise 4.9.) Moreover, from (3.3.1) we see that we may assume that $u(t, \omega)$ and $v(t, \omega)$ are elementary functions. Using Taylor's theorem we get

$$\begin{aligned} g(t, X_t) &= g(0, X_0) + \sum_j \Delta g(t_j, X_j) = g(0, X_0) + \sum_j \frac{\partial g}{\partial t} \Delta t_j + \sum_j \frac{\partial g}{\partial x} \Delta X_j \\ &\quad + \frac{1}{2} \sum_j \frac{\partial^2 g}{\partial t^2} (\Delta t_j)^2 + \sum_j \frac{\partial^2 g}{\partial t \partial x} (\Delta t_j)(\Delta X_j) + \frac{1}{2} \sum_j \frac{\partial^2 g}{\partial x^2} (\Delta X_j)^2 + \sum_j R_j , \end{aligned}$$

where $\frac{\partial g}{\partial t}$, $\frac{\partial g}{\partial x}$ etc. are evaluated at the points (t_j, X_{t_j}) ,

$$\Delta t_j = t_{j+1} - t_j, \quad \Delta X_j = X_{t_{j+1}} - X_{t_j}, \quad \Delta g(t_j, X_j) = g(t_{j+1}, X_{t_{j+1}}) - g(t_j, X_j)$$

and $R_j = o(|\Delta t_j|^2 + |\Delta X_j|^2)$ for all j .

If $\Delta t_j \rightarrow 0$ then

$$\sum_j \frac{\partial g}{\partial t} \Delta t_j = \sum_j \frac{\partial g}{\partial t}(t_j, X_j) \Delta t_j \rightarrow \int_0^t \frac{\partial g}{\partial t}(s, X_s) ds \quad (4.1.10)$$

$$\sum_j \frac{\partial g}{\partial x} \Delta X_j = \sum_j \frac{\partial g}{\partial x}(t_j, X_j) \Delta X_j \rightarrow \int_0^t \frac{\partial g}{\partial x}(s, X_s) dX_s . \quad (4.1.11)$$

Moreover, since u and v are elementary we get

$$\begin{aligned} \sum_j \frac{\partial^2 g}{\partial x^2} (\Delta X_j)^2 &= \sum_j \frac{\partial^2 g}{\partial x^2} u_j^2 (\Delta t_j)^2 + 2 \sum_j \frac{\partial^2 g}{\partial x^2} u_j v_j (\Delta t_j) (\Delta B_j) \\ &+ \sum_j \frac{\partial^2 g}{\partial x^2} v_j^2 \cdot (\Delta B_j)^2, \quad \text{where } u_j = u(t_j, \omega), v_j = v(t_j, \omega) . \end{aligned} \quad (4.1.12)$$

The first two terms here tend to 0 as $\Delta t_j \rightarrow 0$. For example,

$$\begin{aligned} E \left[\left(\sum_j \frac{\partial^2 g}{\partial x^2} u_j v_j (\Delta t_j) (\Delta B_j) \right)^2 \right] &= \\ &= \sum_j E \left[\left(\frac{\partial^2 g}{\partial x^2} u_j v_j \right)^2 \right] (\Delta t_j)^3 \rightarrow 0 \quad \text{as } \Delta t_j \rightarrow 0 . \end{aligned}$$

We claim that the last term tends to

$$\int_0^t \frac{\partial^2 g}{\partial x^2} v^2 ds \quad \text{in } L^2(P), \text{ as } \Delta t_j \rightarrow 0 .$$

To prove this, put $a(t) = \frac{\partial^2 g}{\partial x^2}(t, X_t) v^2(t, \omega)$, $a_j = a(t_j)$ and consider

$$E \left[\left(\sum_j a_j (\Delta B_j)^2 - \sum_j a_j \Delta t_j \right)^2 \right] = \sum_{i,j} E[a_i a_j ((\Delta B_i)^2 - \Delta t_i)((\Delta B_j)^2 - \Delta t_j)] .$$

If $i < j$ then $a_i a_j ((\Delta B_i)^2 - \Delta t_i)$ and $(\Delta B_j)^2 - \Delta t_j$ are independent so the terms vanish in this case, and similarly if $i > j$. So we are left with

$$\begin{aligned} \sum_j E[a_j^2 ((\Delta B_j)^2 - \Delta t_j)^2] &= \sum_j E[a_j^2] \cdot E[(\Delta B_j)^4 - 2(\Delta B_j)^2 \Delta t_j + (\Delta t_j)^2] \\ &= \sum_j E[a_j^2] \cdot (3(\Delta t_j)^2 - 2(\Delta t_j)^2 + (\Delta t_j)^2) = 2 \sum_j E[a_j^2] \cdot (\Delta t_j)^2 \\ &\rightarrow 0 \quad \text{as } \Delta t_j \rightarrow 0 . \end{aligned}$$

In other words, we have established that

$$\sum_j a_j (\Delta B_j)^2 \rightarrow \int_0^t a(s) ds \quad \text{in } L^2(P) \text{ as } \Delta t_j \rightarrow 0$$

and this is often expressed shortly by the striking formula

$$(dB_t)^2 = dt. \quad (4.1.13)$$

The argument above also proves that $\sum R_j \rightarrow 0$ as $\Delta t_j \rightarrow 0$. That completes the proof of the Itô formula. $\square$

Remark. Note that it is enough that $g(t, x)$ is C^2 on $[0, \infty) \times U$, if $U \subset \mathbf{R}$ is an open set such that $X_t(\omega) \in U$ for all $t \geq 0, \omega \in \Omega$. Moreover, it is sufficient that $g(t, x)$ is C^1 w.r.t. t and C^2 w.r.t. x .

4.2 The Multi-dimensional Itô Formula

We now turn to the situation in higher dimensions: Let $B(t, \omega) = (B_1(t, \omega), \dots, B_m(t, \omega))$ denote m -dimensional Brownian motion. If each of the processes $u_i(t, \omega)$ and $v_{ij}(t, \omega)$ satisfies the conditions given in Definition 4.1.1 ($1 \leq i \leq n$, $1 \leq j \leq m$) then we can form the following n Itô processes

$$\begin{cases} dX_1 = u_1 dt + v_{11} dB_1 + \dots + v_{1m} dB_m \\ \vdots \\ dX_n = u_n dt + v_{n1} dB_1 + \dots + v_{nm} dB_m \end{cases} \quad (4.2.1)$$

Or, in matrix notation simply

$$dX(t) = u dt + v dB(t), \quad (4.2.2)$$

where

$$X(t) = \begin{pmatrix} X_1(t) \\ \vdots \\ X_n(t) \end{pmatrix}, \quad u = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_{11} & \dots & v_{1m} \\ \vdots & & \vdots \\ v_{n1} & \dots & v_{nm} \end{pmatrix}, \quad dB(t) = \begin{pmatrix} dB_1(t) \\ \vdots \\ dB_m(t) \end{pmatrix} \quad (4.2.3)$$

Such a process $X(t)$ is called an **n -dimensional Itô process** (or just an Itô process).

We now ask: What is the result of applying a smooth function to X ? The answer is given by

Theorem 4.2.1 (The general Itô formula).

Let

$$dX(t) = u dt + v dB(t)$$

be an n -dimensional Itô process as above. Let $g(t, x) = (g_1(t, x), \dots, g_p(t, x))$ be a C^2 map from $[0, \infty) \times \mathbf{R}^n$ into $\mathbf{R}^p$. Then the process

$$Y(t, \omega) = g(t, X(t))$$

is again an Itô process, whose component number k, Y_k , is given by

$$dY_k = \frac{\partial g_k}{\partial t}(t, X)dt + \sum_i \frac{\partial g_k}{\partial x_i}(t, X)dX_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 g_k}{\partial x_i \partial x_j}(t, X)dX_i dX_j$$

where $dB_i dB_j = \delta_{ij}dt$, $dB_i dt = dt dB_i = 0$.

The proof is similar to the 1-dimensional version (Theorem 4.1.2) and is omitted.

Example 4.2.2. Let $B = (B_1, \dots, B_n)$ be Brownian motion in $\mathbf{R}^n$, $n \geq 2$, and consider

$$R(t, \omega) = |B(t, \omega)| = (B_1^2(t, \omega) + \dots + B_n^2(t, \omega))^{\frac{1}{2}},$$

i.e. the distance to the origin of $B(t, \omega)$. The function $g(t, x) = |x|$ is not C^2 at the origin, but since B_t never hits the origin, a.s. when $n \geq 2$ (see Exercise 9.7) Itô's formula still works and we get

$$dR = \sum_{i=1}^n \frac{B_i dB_i}{R} + \frac{n-1}{2R} dt.$$

The process R is called the *n-dimensional Bessel process* because its generator (Chapter 7) is the Bessel differential operator $Af(x) = \frac{1}{2}f''(x) + \frac{n-1}{2x}f'(x)$. See Example 8.4.1.

4.3 The Martingale Representation Theorem

Let $B(t) = (B_1(t), \dots, B_n(t))$ be n -dimensional Brownian motion. In Chapter 3 (Corollary 3.2.6) we proved that if $v \in \mathcal{V}^n$ then the Itô integral

$$X_t = X_0 + \int_0^t v(s, \omega) dB(s); \quad t \geq 0$$

is always a martingale w.r.t. filtration $\mathcal{F}_t^{(n)}$ (and w.r.t. the probability measure P). In this section we will prove that the converse is also true: Any $\mathcal{F}_t^{(n)}$ -martingale (w.r.t. P) can be represented as an Itô integral. This result, called the *martingale representation theorem*, is important for many applications, for example in mathematical finance. See Chapter 12. For simplicity we prove the result only when $n = 1$, but the reader can easily verify that essentially the same proof works for arbitrary n .

We first establish some auxiliary results.

Lemma 4.3.1. *Fix $T > 0$. The set of random variables*

$$\{\phi(B_{t_1}, \dots, B_{t_n}); t_i \in [0, T], \phi \in C_0^\infty(\mathbf{R}^n), n = 1, 2, \dots\}$$

is dense in $L^2(\mathcal{F}_T, P)$.

Proof. Let $\{t_i\}_{i=1}^\infty$ be a dense subset of $[0, T]$ and for each $n = 1, 2, \dots$ let $\mathcal{H}_n$ be the σ -algebra generated by $B_{t_1}(\cdot), \dots, B_{t_n}(\cdot)$. Then clearly

$$\mathcal{H}_n \subset \mathcal{H}_{n+1}$$

and $\mathcal{F}_T$ is the smallest σ -algebra containing all the $\mathcal{H}_n$'s. Choose $g \in L^2(\mathcal{F}_T, P)$. Then by the martingale convergence theorem Corollary C.9 (Appendix C) we have that

$$g = E[g|\mathcal{F}_T] = \lim_{n \rightarrow \infty} E[g|\mathcal{H}_n].$$

The limit is pointwise a.e. (P) and in $L^2(\mathcal{F}_T, P)$. By the Doob-Dynkin Lemma (Lemma 2.1.2) we can write, for each n ,

$$E[g|\mathcal{H}_n] = g_n(B_{t_1}, \dots, B_{t_n})$$

for some Borel measurable function $g_n: \mathbf{R}^n \rightarrow \mathbf{R}$. Each such $g_n(B_{t_1}, \dots, B_{t_n})$ can be approximated in $L^2(\mathcal{F}_T, P)$ by functions $\phi_n(B_{t_1}, \dots, B_{t_n})$ where $\phi_n \in C_0^\infty(\mathbf{R}^n)$ and the result follows. $\square$

Lemma 4.3.2. *The linear span of random variables of the type*

$$\exp \left\{ \int_0^T h(t) dB_t(\omega) - \frac{1}{2} \int_0^T h^2(t) dt \right\}; \quad h \in L^2[0, T] \text{ (deterministic)} \quad (4.3.1)$$

is dense in $L^2(\mathcal{F}_T, P)$.

Proof. Suppose $g \in L^2(\mathcal{F}_T, P)$ is orthogonal (in $L^2(\mathcal{F}_T, P)$) to all functions of the form (4.3.1). Then in particular

$$G(\lambda) := \int_{\Omega} \exp\{\lambda_1 B_{t_1}(\omega) + \dots + \lambda_n B_{t_n}(\omega)\} g(\omega) dP(\omega) = 0 \quad (4.3.2)$$

for all $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbf{R}^n$ and all $t_1, \dots, t_n \in [0, T]$. The function $G(\lambda)$ is real analytic in $\lambda \in \mathbf{R}^n$ and hence G has an analytic extension to the complex space $\mathbf{C}^n$ given by

$$G(z) = \int_{\Omega} \exp\{z_1 B_{t_1}(\omega) + \dots + z_n B_{t_n}(\omega)\} g(\omega) dP(\omega) \quad (4.3.3)$$

for all $z = (z_1, \dots, z_n) \in \mathbf{C}^n$. (See the estimates in Exercise 2.8 b.) Since $G = 0$ on $\mathbf{R}^n$ and G is analytic, $G = 0$ on $\mathbf{C}^n$. In particular, $G(iy_1, iy_2, \dots, iy_n) = 0$ for all $y = (y_1, \dots, y_n) \in \mathbf{R}^n$. But then we get, for $\phi \in C_0^\infty(\mathbf{R}^n)$,

$$\begin{aligned}
& \int_{\Omega} \phi(B_{t_1}, \dots, B_{t_n}) g(\omega) dP(\omega) \\
&= \int_{\Omega} (2\pi)^{-n/2} \left(\int_{\mathbf{R}^n} \hat{\phi}(y) e^{i(y_1 B_{t_1} + \dots + y_n B_{t_n})} dy \right) g(\omega) dP(\omega) \\
&= (2\pi)^{-n/2} \int_{\mathbf{R}^n} \hat{\phi}(y) \left(\int_{\Omega} e^{i(y_1 B_{t_1} + \dots + y_n B_{t_n})} g(\omega) dP(\omega) \right) dy \\
&= (2\pi)^{-n/2} \int_{\mathbf{R}^n} \hat{\phi}(y) G(iy) dy = 0, \tag{4.3.4}
\end{aligned}$$

where

$$\hat{\phi}(y) = (2\pi)^{-n/2} \int_{\mathbf{R}^n} \phi(x) e^{-i x \cdot y} dx$$

is the Fourier transform of ϕ and we have used the inverse Fourier transform theorem

$$\phi(x) = (2\pi)^{-n/2} \int_{\mathbf{R}^n} \hat{\phi}(y) e^{i x \cdot y} dy$$

(see e.g. Folland (1984)).

By (4.3.4) and Lemma 4.3.1 g is orthogonal to a dense subset of $L^2(\mathcal{F}_T, P)$ and we conclude that $g = 0$. Therefore the linear span of the functions in (4.3.1) must be dense in $L^2(\mathcal{F}_T, P)$ as claimed. $\square$

Suppose $B(t) = (B_1(t), \dots, B_n(t))$ is n -dimensional. If $v(s, \omega) \in \mathcal{V}^n(0, T)$ then the random variable

$$V(\omega) := \int_0^T v(t, \omega) dB(t) \tag{4.3.5}$$

is $\mathcal{F}_T^{(n)}$ -measurable and by the Itô isometry

$$E[V^2] = \int_0^T E[v^2(t, \cdot)] dt < \infty, \quad \text{so } V \in L^2(\mathcal{F}_T^{(n)}, P).$$

The next result states that any $F \in L^2(\mathcal{F}_T^{(n)}, P)$ can be represented this way:

Theorem 4.3.3 (The Itô representation theorem).

Let $F \in L^2(\mathcal{F}_T^{(n)}, P)$. Then there exists a unique stochastic process $f(t, \omega) \in \mathcal{V}^n(0, T)$ such that

$$F(\omega) = E[F] + \int_0^T f(t, \omega) dB(t) . \quad (4.3.6)$$

Proof. Again we consider only the case $n = 1$. (The proof in the general case is similar.) First assume that F has the form (4.3.1), i.e.

$$F(\omega) = \exp \left\{ \int_0^T h(t) dB_t(\omega) - \frac{1}{2} \int_0^T h^2(t) dt \right\}$$

for some $h(t) \in L^2[0, T]$.

Define

$$Y_t(\omega) = \exp \left\{ \int_0^t h(s) dB_s(\omega) - \frac{1}{2} \int_0^t h^2(s) ds \right\} ; \quad 0 \leq t \leq T .$$

Then by Itô's formula

$$dY_t = Y_t(h(t)dB_t - \frac{1}{2}h^2(t)dt) + \frac{1}{2}Y_t(h(t)dB_t)^2 = Y_t h(t)dB_t$$

so that

$$Y_t = 1 + \int_0^t Y_s h(s) dB_s ; \quad t \in [0, T] .$$

Therefore

$$F = Y_T = 1 + \int_0^T Y_s h(s) dB_s$$

and hence $E[F] = 1$. So (4.3.6) holds in this case. By linearity (4.3.6) also holds for linear combinations of functions of the form (4.3.1). So if $F \in L^2(\mathcal{F}_T, P)$ is arbitrary, we approximate F in $L^2(\mathcal{F}_T, P)$ by linear combinations F_n of functions of the form (4.3.1). Then for each n we have

$$F_n(\omega) = E[F_n] + \int_0^T f_n(s, \omega) dB_s(\omega), \quad \text{where } f_n \in \mathcal{V}(0, T) .$$

By the Itô isometry

$$\begin{aligned} E[(F_n - F_m)^2] &= E \left[(E[F_n - F_m] + \int_0^T (f_n - f_m) dB)^2 \right] \\ &= (E[F_n - F_m])^2 + \int_0^T E[(f_n - f_m)^2] dt \rightarrow 0 \quad \text{as } n, m \rightarrow \infty \end{aligned}$$

so $\{f_n\}$ is a Cauchy sequence in $L^2([0, T] \times \Omega)$ and hence converges to some $f \in L^2([0, T] \times \Omega)$. Since $f_n \in \mathcal{V}(0, T)$ we have $f \in \mathcal{V}(0, T)$. (A subsequence of $\{f_n(t, \omega)\}$ converges to $f(t, \omega)$ for a.a. $(t, \omega) \in [0, T] \times \Omega$. Therefore $f(t, \cdot)$ is $\mathcal{F}_t$ -measurable for a.a. t . So by modifying $f(t, \omega)$ on a t -set of measure 0 we can obtain that $f(t, \omega)$ is $\mathcal{F}_t$ -adapted.) Again using the Itô isometry we see that

$$F = \lim_{n \rightarrow \infty} F_n = \lim_{n \rightarrow \infty} \left(E[F_n] + \int_0^T f_n dB \right) = E[F] + \int_0^T f dB ,$$

the limit being taken in $L^2(\mathcal{F}_T, P)$. Hence the representation (4.3.6) holds for all $F \in L^2(\mathcal{F}_T, P)$.

The uniqueness follows from the Itô isometry: Suppose

$$F(\omega) = E[F] + \int_0^T f_1(t, \omega) dB_t(\omega) = E[F] + \int_0^T f_2(t, \omega) dB_t(\omega)$$

with $f_1, f_2 \in \mathcal{V}(0, T)$. Then

$$0 = E\left[\left(\int_0^T (f_1(t, \omega) - f_2(t, \omega)) dB_t(\omega)\right)^2\right] = \int_0^T E[(f_1(t, \omega) - f_2(t, \omega))^2] dt$$

and therefore $f_1(t, \omega) = f_2(t, \omega)$ for a.a. $(t, \omega) \in [0, T] \times \Omega$. $\square$

Remark. The process $f(t, \omega)$ can be expressed in terms of the Frechet derivative and also in terms of the Malliavin derivative of $F(\omega)$. See Clark (1970/71) and Ocone (1984).

Theorem 4.3.4 (The martingale representation theorem).

Let $B(t) = (B_1(t), \dots, B_n(t))$ be n -dimensional. Suppose M_t is an $\mathcal{F}_t^{(n)}$ -martingale (w.r.t. P) and that $M_t \in L^2(P)$ for all $t \geq 0$. Then there exists a unique stochastic process $g(s, \omega)$ such that $g \in \mathcal{V}^{(n)}(0, t)$ for all $t \geq 0$ and

$$M_t(\omega) = E[M_0] + \int_0^t g(s, \omega) dB(s) \quad \text{a.s., for all } t \geq 0 .$$

Proof ($n = 1$). By Theorem 4.3.3 applied to $T = t$, $F = M_t$, we have that for all t there exists a unique $f^{(t)}(s, \omega) \in L^2(\mathcal{F}_t, P)$ such that

$$M_t(\omega) = E[M_t] + \int_0^t f^{(t)}(s, \omega) dB_s(\omega) = E[M_0] + \int_0^t f^{(t)}(s, \omega) dB_s(\omega) .$$

Now assume $0 \leq t_1 < t_2$. Then

$$\begin{aligned}
M_{t_1} &= E[M_{t_2} | \mathcal{F}_{t_1}] = E[M_0] + E\left[\int_0^{t_2} f^{(t_2)}(s, \omega) dB_s(\omega) | \mathcal{F}_{t_1}\right] \\
&= E[M_0] + \int_0^{t_1} f^{(t_2)}(s, \omega) dB_s(\omega) .
\end{aligned} \tag{4.3.7}$$

But we also have

$$M_{t_1} = E[M_0] + \int_0^{t_1} f^{(t_1)}(s, \omega) dB_s(\omega) . \tag{4.3.8}$$

Hence, comparing (4.3.7) and (4.3.8) we get that

$$0 = E\left[\left(\int_0^{t_1} (f^{(t_2)} - f^{(t_1)}) dB\right)^2\right] = \int_0^{t_1} E[(f^{(t_2)} - f^{(t_1)})^2] ds$$

and therefore

$$f^{(t_1)}(s, \omega) = f^{(t_2)}(s, \omega) \quad \text{for a.a. } (s, \omega) \in [0, t_1] \times \Omega .$$

So we can define $f(s, \omega)$ for a.a. $s \in [0, \infty) \times \Omega$ by setting

$$f(s, \omega) = f^{(N)}(s, \omega) \quad \text{if } s \in [0, N]$$

and then we get

$$M_t = E[M_0] + \int_0^t f^{(t)}(s, \omega) dB_s(\omega) = E[M_0] + \int_0^t f(s, \omega) dB_s(\omega) \quad \text{for all } t \geq 0 .$$

□

Exercises

4.1. Use Itô's formula to write the following stochastic processes X_t on the standard form

$$dX_t = u(t, \omega)dt + v(t, \omega)dB_t$$

for suitable choices of $u \in \mathbf{R}^n$, $v \in \mathbf{R}^{n \times m}$ and dimensions n, m :

- a) $X_t = B_t^2$, where B_t is 1-dimensional
- b) $X_t = 2 + t + e^{B_t}$ (B_t is 1-dimensional)
- c) $X_t = B_1^2(t) + B_2^2(t)$ where (B_1, B_2) is 2-dimensional
- d) $X_t = (t_0 + t, B_t)$ (B_t is 1-dimensional)
- e) $X_t = (B_1(t) + B_2(t) + B_3(t), B_2^2(t) - B_1(t)B_3(t))$, where (B_1, B_2, B_3) is 3-dimensional.

4.2. Use Itô's formula to prove that

$$\int_0^t B_s^2 dB_s = \frac{1}{3} B_t^3 - \int_0^t B_s ds .$$

4.3. Let X_t, Y_t be Itô processes in $\mathbf{R}$. Prove that

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t \cdot dY_t .$$

Deduce the following general *integration by parts formula*

$$\int_0^t X_s dY_s = X_t Y_t - X_0 Y_0 - \int_0^t Y_s dX_s - \int_0^t dX_s \cdot dY_s .$$

4.4. (Exponential martingales)

Suppose $\theta(t, \omega) = (\theta_1(t, \omega), \dots, \theta_n(t, \omega)) \in \mathbf{R}^n$ with $\theta_k(t, \omega) \in \mathcal{V}[0, T]$ for $k = 1, \dots, n$, where $T \leq \infty$. Define

$$Z_t = \exp \left\{ \int_0^t \theta(s, \omega) dB(s) - \frac{1}{2} \int_0^t \theta^2(s, \omega) ds \right\}; \quad 0 \leq t \leq T$$

where $B(s) \in \mathbf{R}^n$ and $\theta^2 = \theta \cdot \theta$ (dot product).

a) Use Itô's formula to prove that

$$dZ_t = Z_t \theta(t, \omega) dB(t) .$$

b) Deduce that Z_t is a martingale for $t \leq T$, provided that

$$Z_t \theta_k(t, \omega) \in \mathcal{V}[0, T] \quad \text{for } 1 \leq k \leq n .$$

Remark. A sufficient condition that Z_t be a martingale is the *Kazamaki condition*

$$E \left[\exp \left(\frac{1}{2} \int_0^t \theta(s, \omega) dB(s) \right) \right] < \infty \quad \text{for all } t \leq T . \quad (4.3.9)$$

This is implied by the following (stronger) *Novikov condition*

$$E \left[\exp \left(\frac{1}{2} \int_0^T \theta^2(s, \omega) ds \right) \right] < \infty . \quad (4.3.10)$$

See e.g. Ikeda & Watanabe (1989), Section III.5, and the references therein.

4.5. Let $B_t \in \mathbf{R}$, $B_0 = 0$. Define

$$\beta_k(t) = E[B_t^k] ; \quad k = 0, 1, 2, \dots ; \quad t \geq 0 .$$

Use Itô's formula to prove that

$$\beta_k(t) = \frac{1}{2}k(k-1) \int_0^t \beta_{k-2}(s) ds ; \quad k \geq 2 .$$

Deduce that

$$E[B_t^4] = 3t^2 \quad (\text{see (2.2.14)})$$

and find

$$E[B_t^6] .$$

4.6. a) For c, α constants, $B_t \in \mathbf{R}$ define

$$X_t = e^{ct + \alpha B_t} .$$

Prove that

$$dX_t = (c + \frac{1}{2}\alpha^2)X_t dt + \alpha X_t dB_t .$$

b) For $c, \alpha_1, \dots, \alpha_n$ constants, $B_t = (B_1(t), \dots, B_n(t)) \in \mathbf{R}^n$ define

$$X_t = \exp \left(ct + \sum_{j=1}^n \alpha_j B_j(t) \right) .$$

Prove that

$$dX_t = \left(c + \frac{1}{2} \sum_{j=1}^n \alpha_j^2 \right) X_t dt + X_t \left(\sum_{j=1}^n \alpha_j dB_j \right) .$$

4.7. Let X_t be an Itô integral

$$dX_t = v(t, \omega) dB_t(\omega) \quad \text{where } v \in \mathbf{R}^n, v \in \mathcal{V}(0, T), B_t \in \mathbf{R}^n, 0 \leq t \leq T .$$

- a) Give an example to show that X_t^2 is not in general a martingale.
- b) Prove that if v is bounded then

$$M_t := X_t^2 - \int_0^t |v_s|^2 ds \quad \text{is a martingale .}$$

The process $\langle X, X \rangle_t := \int_0^t |v_s|^2 ds$ is called the *quadratic variation process* of the martingale X_t . For general processes X_t it is defined by

$$\langle X, X \rangle_t = \lim_{\Delta t_k \rightarrow 0} \sum_{t_k \leq t} |X_{t_{k+1}} - X_{t_k}|^2 \quad (\text{limit in probability}) \quad (4.3.11)$$

where $0 = t_1 < t_2 < \dots < t_n = t$ and $\Delta t_k = t_{k+1} - t_k$. The limit can be shown to exist for continuous square integrable martingales X_t . See e.g. Karatzas and Shreve (1991).

- 4.8.** a) Let B_t denote n -dimensional Brownian motion and let $f: \mathbf{R}^n \rightarrow \mathbf{R}$ be C^2 . Use Itô's formula to prove that

$$f(B_t) = f(B_0) + \int_0^t \nabla f(B_s) dB_s + \frac{1}{2} \int_0^t \Delta f(B_s) ds,$$

where $\Delta = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$ is the Laplace operator.

- b) Assume that $g: \mathbf{R} \rightarrow \mathbf{R}$ is C^1 everywhere and C^2 outside finitely many points $z_1, \dots, z_N$ with $|g''(x)| \leq M$ for $x \notin \{z_1, \dots, z_N\}$. Let B_t be 1-dimensional Brownian motion. Prove that the 1-dimensional version of a) still holds, i.e.

$$g(B_t) = g(B_0) + \int_0^t g'(B_s) dB_s + \frac{1}{2} \int_0^t g''(B_s) ds.$$

(Hint: Choose $f_k \in C^2(\mathbf{R})$ s.t. $f_k \rightarrow g$ uniformly, $f'_k \rightarrow g'$ uniformly and $|f''_k| \leq M$, $f''_k \rightarrow g''$ outside $z_1, \dots, z_N$. Apply a) to f_k and let $k \rightarrow \infty$).

- 4.9.** Prove that we may assume that g and its first two derivatives are bounded in the proof of the Itô formula (Theorem 4.1.2) by proceeding as follows: For fixed $t \geq 0$ and $n = 1, 2, \dots$ choose g_n as in the statement such that $g_n(s, x) = g(s, x)$ for all $s \leq t$ and all $|x| \leq n$. Suppose we have proved that (4.1.9) holds for each g_n . Define the stochastic time

$$\tau_n = \tau_n(\omega) = \inf\{s > 0; |X_s(\omega)| \geq n\}$$

(τ_n is called a *stopping* time (See Chapter 7)) and prove that

$$\left(\int_0^t v \frac{\partial g_n}{\partial x}(s, X_s) \mathcal{X}_{s \leq \tau_n} dB_s := \right) \\ \int_0^{t \wedge \tau_n} v \frac{\partial g_n}{\partial x}(s, X_s) dB_s = \int_0^{t \wedge \tau_n} v \frac{\partial g}{\partial x}(s, X_s) dB_s$$

for each n . This gives that

$$g(t \wedge \tau_n, X_{t \wedge \tau_n}) = g(0, X_0) + \int_0^{t \wedge \tau_n} \left(\frac{\partial g}{\partial s} + u \frac{\partial g}{\partial x} + \frac{1}{2} v^2 \frac{\partial^2 g}{\partial x^2} \right) ds + \int_0^{t \wedge \tau_n} v \frac{\partial g}{\partial x} dB_s$$

and since

$$P[\tau_n > t] \rightarrow 1 \quad \text{as } n \rightarrow \infty$$

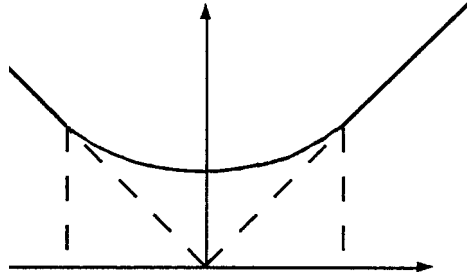
we can conclude that (4.1.9) holds (a.s.) for g .

4.10. (Tanaka's formula and local time).

What happens if we try to apply the Itô formula to $g(B_t)$ when B_t is 1-dimensional and $g(x) = |x|$? In this case g is not C^2 at $x = 0$, so we modify $g(x)$ near $x = 0$ to $g_\epsilon(x)$ as follows:

$$g_\epsilon(x) = \begin{cases} |x| & \text{if } |x| \geq \epsilon \\ \frac{1}{2}(\epsilon + \frac{x^2}{\epsilon}) & \text{if } |x| < \epsilon \end{cases}$$

where $\epsilon > 0$.



a) Apply Exercise 4.8 b) to show that

$$g_\epsilon(B_t) = g_\epsilon(B_0) + \int_0^t g'_\epsilon(B_s) dB_s + \frac{1}{2\epsilon} \cdot |\{s \in [0, t]; B_s \in (-\epsilon, \epsilon)\}|$$

where $|F|$ denotes the Lebesgue measure of the set F .

b) Prove that

$$\int_0^t g'_\epsilon(B_s) \cdot \mathcal{X}_{B_s \in (-\epsilon, \epsilon)} dB_s = \int_0^t \frac{B_s}{\epsilon} \cdot \mathcal{X}_{B_s \in (-\epsilon, \epsilon)} dB_s \rightarrow 0$$

in $L^2(P)$ as $\epsilon \rightarrow 0$.

(Hint: Apply the Itô isometry to

$$E \left[\left(\int_0^t \frac{B_s}{\epsilon} \cdot \mathcal{X}_{B_s \in (-\epsilon, \epsilon)} dB_s \right)^2 \right].$$

c) By letting $\epsilon \rightarrow 0$ prove that

$$|B_t| = |B_0| + \int_0^t \text{sign}(B_s) dB_s + L_t(\omega) , \quad (4.3.12)$$

where

$$L_t = \lim_{\epsilon \rightarrow 0} \frac{1}{2\epsilon} \cdot |\{s \in [0, t]; B_s \in (-\epsilon, \epsilon)\}| \quad (\text{limit in } L^2(P))$$

and

$$\text{sign}(x) = \begin{cases} -1 & \text{for } x \leq 0 \\ 1 & \text{for } x > 0 \end{cases} .$$

L_t is called the *local time* for Brownian motion at 0 and (4.3.12) is the *Tanaka formula* (for Brownian motion). (See e.g. Rogers and Williams (1987)).

4.11. Use Itô's formula (for example in the form of Exercise 4.3) to prove that the following stochastic processes are $\{\mathcal{F}_t\}$ -martingales:

- a) $X_t = e^{\frac{1}{2}t} \cos B_t \quad (B_t \in \mathbf{R})$
- b) $X_t = e^{\frac{1}{2}t} \sin B_t \quad (B_t \in \mathbf{R})$
- c) $X_t = (B_t + t) \exp(-B_t - \frac{1}{2}t) \quad (B_t \in \mathbf{R})$.

4.12. Let $dX_t = u(t, \omega)dt + v(t, \omega)dB_t$ be an Itô process in $\mathbf{R}^n$ such that

$$E \left[\int_0^t |u(r, \omega)| dr \right] + E \left[\int_0^t |vv^T(r, \omega)| dr \right] < \infty \quad \text{for all } t \geq 0 .$$

Suppose X_t is an $\{\mathcal{F}_t^{(n)}\}$ -martingale. Prove that

$$u(s, \omega) = 0 \quad \text{for a.a. } (s, \omega) \in [0, \infty) \times \Omega . \quad (4.3.13)$$

Remarks:

- 0) 1) This result may be regarded as a special case of the Martingale Representation Theorem.
- 2) The conclusion (4.3.13) does not hold if the filtration $\mathcal{F}_t^{(n)}$ is replaced by the σ -algebras $\mathcal{M}_t$ generated by $X_s(\cdot)$; $s \leq t$, i.e. if we only assume that X_t is a martingale w.r.t. its own filtration. See e.g. the Brownian motion characterization in Chapter 8.

Hint for the solution:

If X_t is an $\mathcal{F}_t^{(n)}$ -martingale, then deduce that

$$E \left[\int_t^s u(r, \omega) dr | \mathcal{F}_t^{(n)} \right] = 0 \quad \text{for all } s \geq t .$$

Differentiate w.r.t. s to deduce that

$$E[u(s, \omega) | \mathcal{F}_t^{(n)}] = 0 \quad \text{a.s., for a.a. } s > t.$$

Then let $t \uparrow s$ and apply Corollary C.9.

- 4.13.** Let $dX_t = u(t, \omega)dt + dB_t$ ($u \in \mathbf{R}$, $B_t \in \mathbf{R}$) be an Itô process and assume for simplicity that u is bounded. Then from Exercise 4.12 we know that unless $u = 0$ the process X_t is not an $\mathcal{F}_t$ -martingale. However, it turns out that we can construct an $\mathcal{F}_t$ -martingale from X_t by multiplying by a suitable exponential martingale. More precisely, define

$$Y_t = X_t M_t$$

where

$$M_t = \exp \left(- \int_0^t u(r, \omega) dB_r - \frac{1}{2} \int_0^t u^2(r, \omega) dr \right).$$

Use Itô's formula to prove that

$$Y_t \text{ is an } \mathcal{F}_t\text{-martingale.}$$

Remarks:

- a) a) Compare with Exercise 4.11 c).
 b) This result is a special case of the important *Girsanov Theorem*. It can be interpreted as follows: $\{X_t\}_{t \leq T}$ is a martingale w.r.t the measure Q defined on $\mathcal{F}_T$ by

$$dQ = M_T dP \quad (T < \infty).$$

See Section 8.6.

- 4.14.** In each of the cases below find the process $f(t, \omega) \in \mathcal{V}[0, T]$ such that (4.3.6) holds, i.e.

$$F(\omega) = E[F] + \int_0^T f(t, \omega) dB_t(\omega).$$

- a) $F(\omega) = B_T(\omega)$ b) $F(\omega) = \int_0^T B_t(\omega) dt$
 c) $F(\omega) = B_T^2(\omega)$ d) $F(\omega) = B_T^3(\omega)$
 e) $F(\omega) = e^{B_T(\omega)}$ f) $F(\omega) = \sin B_T(\omega)$

- 4.15.** Let $x > 0$ be a constant and define

$$X_t = (x^{1/3} + \frac{1}{3}B_t)^3; \quad t \geq 0.$$

Show that

$$dX_t = \frac{1}{3}X_t^{1/3}dt + X_t^{2/3}dB_t; \quad X_0 = x.$$